VELADIAL

Document 13 — Firmware Prep Validation Plan

Firmware Prep & Validation Plan

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Status: Validation plan — hardware must be validated before production firmware
Date: 2026-05-22
Project: VelaDial

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1. Purpose

This document defines the controlled validation sequence required before writing production ESPHome firmware for VelaDial.

This is not production firmware. This plan validates hardware, ESPHome component support, I2C addresses, and the Home Assistant control path before any production YAML is written.

Current YAML file status (updated 2026-05-29): - `esphome/door_side_rotary.yaml` — **Production multi-theme firmware engine** (20 selectable UI themes, compile-passing, PR #56). Hardware validation NOT YET TESTED on physical board. - `esphome/bedside_gesture.yaml` — basic APDS-9960 bring-up with known issues (60s update interval too slow for gestures)

This document supports `docs/06_Implementation_Plan.md` by providing a practical validation checklist. Production firmware requires validated hardware, confirmed component support, and verified I2C communication.

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2. Validation Principles

- **Validate one subsystem at a time.** Do not combine display validation with sensor validation with HA integration in a single step.
- **Record pass/fail results.** Every validation step must produce a written result document.
- **Do not build full production UI until display/touch/encoder/backlight are validated.** The 3-page LVGL UI (Power, Brightness, Presets) comes after hardware validation.

- **Do not implement sensor fusion in v1.** APDS-9960 standalone gestures and VL53L4CD standalone hand-hold are v1. Fusion is v2/future only.
- **Do not write VL53L4CD hold-nightlight firmware until support path is verified.** VL53L4CD support verification must complete before bedside ToF firmware work.
- **Do not switch to VL53L0X unless VL53L4CD validation blocks progress and Hardik approves.** VL53L0X is verified fallback only, not selected default.

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3. Validation Checklist

This checklist supports the phases in `docs/06_Implementation_Plan.md`. Complete each item and record results in `hardware/validation_results.md`.

A. VL53L4CD Support Verification

Goal: Determine if VL53L4CD has a viable ESPHome support path.

Actions:

1. Check official ESPHome documentation at https://esphome.io/components/sensor/ for `vl53l4cd`
2. Check ESPHome GitHub repository for `vl53l4cd` component
3. Search trusted community/external components
4. **Decide one path:**
5. VL53L4CD native support
6. Maintained community component
7. Custom C++ component
8. Defer hold-nightlight (APDS-only bedside)
9. VL53L0X fallback only with owner approval

Pass/Fail Criteria: - ✅ **PASS:** Native support confirmed, community component found, custom path feasible, or decision to defer - ⚠️ **CONDITIONAL PASS:** VL53L0X fallback approved with documented threshold adjustments - ❌ **FAIL:** No support path and no fallback approved

Gate: Bedside ToF validation cannot begin until this completes.

B. ELECROW Board and Pinout Validation

Goal: Verify the physical ELECROW board matches documented pinout.

Actions:

1. Photograph board front/back, capture silkscreen and revision numbers
2. Compare to `hardware/elecrow_pinout.md`, note discrepancies
3. Validate each subsystem:
4. Display: GC9A01A initializes, shows test pattern
5. Touch: CST816 responds at `0x15`, generates coordinates
6. Encoder rotation: GPIO45/42 detect rotation, direction verified
7. Encoder press: GPIO41 detects button press
8. Backlight PWM: GPIO46 controls brightness 0-100%
9. WS2812 LED ring: GPIO48 lights up all 5 LEDs
10. I2C bus: Scan finds touch at `0x15`

Pass/Fail Criteria: - ✅ **PASS:** All subsystems respond correctly, or issues documented with workarounds - ✅ **PASS:** Board revision identified and pinout confirmed - ⚠️ **PARTIAL:** Some GPIOs differ but workarounds documented - ❌ **FAIL:** Critical GPIOs wrong and no workaround

Gate: Door-side sensors cannot be validated until this passes. Production door-side firmware has been written (PR #56) but cannot be validated on hardware until this passes.

C. Door-Side Sensor Validation

Goal: Verify TSL2591 and SHT45 function correctly on door-side I2C bus.

Actions:

1. Wire sensors via STEMMA QT (TSL2591 + SHT45)
2. Run I2C scan: expect `0x29` (TSL2591), `0x44` (SHT45), `0x15` (touch)
3. Test TSL2591: dark (<5 lux), dim (5-50), bright (>50), direct (>500)
4. Test SHT45: compare temp (±2°C) and humidity (±5%) to reference
5. Confirm sensors don't break display/touch/encoder

Pass/Fail Criteria: - ✅ **PASS:** I2C scan finds `0x29` and `0x44`, readings appropriate, no interference - ⚠️ **PARTIAL:** Sensors work with limitations (documented) - ❌ **FAIL:** Sensors not detected or cause instability

Gate: Adaptive backlight firmware cannot be written until this passes.

D. Bedside APDS-9960 Validation

Goal: Verify APDS-9960 gesture detection works reliably in bedside mounting position.

Actions:

1. Wire APDS-9960 via STEMMA QT to bedside ESP32-C6
2. Run I2C scan: expect `0x39`
3. Test proximity sensor (0-255 values)
4. Test gestures in actual mounting: left/right 10 attempts each, document success rate
5. Test environmental conditions: lights on/off, IR interference
6. Fix current issue: change `update_interval` from 60s to 50-100ms or interrupt-driven
7. Tune proximity threshold and gesture sensitivity
8. Add and validate ~1s cooldown after gesture

Pass/Fail Criteria: - ✅ **PASS:** I2C finds `0x39`, left/right gestures ≥80% reliable, cooldown works - ⚠️ **PARTIAL:** Gestures work with lower reliability (document conditions) - ❌ **FAIL:** Gestures <50% reliable or false trigger constantly

Gate: Bedside APDS v1 firmware can be written after this passes. If this fails, consider button fallback or accept reduced reliability.

E. Bedside ToF Validation

Goal: Verify VL53L4CD (or VL53L0X fallback) distance sensing and hand-hold detection.

Prerequisite: VL53L4CD support verification must complete first.

Actions:

1. Wire ToF sensor via STEMMA QT to bedside ESP32-C6
2. Run I2C scan: expect `0x29`
3. Test distance readings at 5cm, 10cm, 30cm, 100cm (±10mm acceptable)
4. If VL53L0X fallback: note 30mm minimum range, re-tune thresholds
5. Test hand-hold: 5-10cm for 1.5s triggers "holding", removal cancels, pass-by <200ms rejected, ~5s cooldown
6. Test for interference with APDS-9960

Pass/Fail Criteria: - ✅ **PASS:** I2C finds `0x29`, readings accurate, hand-hold detection works, no interference - ⚠️ **PARTIAL:** ToF works with limitations (documented) - ❌ **FAIL:** Readings unstable or hand-hold detection unreliable

Gate: Bedside ToF nightlight firmware can be written after this passes. If this fails and VL53L0X fallback not approved, bedside ships APDS-only.

F. Home Assistant Integration Validation

Goal: Verify ESPHome devices control HA lights via local network with acceptable latency.

Actions:

1. Confirm `light.bedroom_group` exists, controls all 5 bulbs
2. Confirm local control with internet disconnected
3. Measure response time: target <500ms
4. Test door-side: rotary → brightness, button → toggle
5. Test bedside APDS: left → off, right → on
6. Test nightlight (only if ToF validated): hand hold → nightlight scene

Pass/Fail Criteria: - ✅ **PASS:** Group controls all bulbs, local control works, latency <500ms, commands reliable - ⚠️ **PARTIAL:** Works but latency 500ms-1s (documented) - ❌ **FAIL:** Commands don't reach bulbs or latency >1s

Gate: Production firmware development can begin after this passes.

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4. Validation YAML Files

Validation YAML files will be created one phase at a time when needed. Do not create them now.

Future validation YAMLs: - `esphome/door_hardware_validation.yaml` — ELECROW display, touch, encoder, backlight, WS2812 - `esphome/door_sensors_validation.yaml` — TSL2591 and SHT45 - `esphome/door_integration_test.yaml` — HA light control from door-side - `esphome/bedside_apds_validation.yaml` — APDS-9960 gesture detection - `esphome/bedside_tof_validation.yaml` — VL53L4CD or VL53L0X distance sensing - `esphome/bedside_integration_test.yaml` — HA light control from bedside

Each YAML will be created only when its validation phase begins.

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5. Risk Ranking

🔴 CRITICAL RISK

VL53L4CD Support Risk - Impact: Bedside nightlight feature may be impossible without hardware swap - Probability: High (official support unverified) - Mitigation: Support verification checklist; VL53L0X fallback documented - Consequence: Nightlight feature blocked if no support path and no fallback approved

🟠 HIGH RISK

ELECROW Pinout Risk - Impact: Wrong GPIO assignments = non-functional hardware - Probability: Medium (multiple board revisions exist) - Mitigation: Physical validation checklist - Consequence: PCB rework or board replacement needed

APDS Gesture Reliability Risk - Impact: Unreliable gestures = poor UX, feature unusable - Probability: Medium (gesture sensors finicky in real environments) - Mitigation: Real-world testing; tune thresholds - Consequence: May need button fallback or accept reduced reliability

🟡 MEDIUM RISK

I2C/Address Risk - Impact: Sensors not detected or conflict - Probability: Low (architecture separates `0x29` devices) - Mitigation: I2C scans in validation checklists - Consequence: Wiring error or defective sensor

Network/HA Control Path Risk - Impact: Commands don't reach bulbs or latency too high - Probability: Low (LocalTuya proven, LAN reliable) - Mitigation: Integration validation checklist - Consequence: Network/HA configuration debugging needed

LVGL Performance Risk - Impact: UI lag, crashes, out-of-memory - Probability: Low (8MB PSRAM sufficient for 240×240) - Mitigation: Monitor heap during development - Consequence: May need UI simplification

🟢 LOW RISK

TSL2591/SHT45 Integration Risk - Impact: Ambient light or temp/humidity not working - Probability: Very Low (native ESPHome support) - Mitigation: Door sensor validation checklist - Consequence: Display won't adapt; diagnostics missing (not critical)

Power Supply Risk - Impact: Brownouts from insufficient power - Probability: Very Low (sensors low power) - Mitigation: Use quality 5V/2A+ supply - Consequence: Intermittent crashes

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6. Go/No-Go Gates

Before door-side production UI work:

- ✅ ELECROW validation passed
- ✅ Door sensors validated (or documented as deferred)
- ✅ HA integration works with acceptable latency

If any gate fails: Debug and re-test before writing production UI. Do not build 3-page LVGL on unvalidated hardware.

Before bedside APDS v1 firmware:

- ✅ APDS validation passed (gestures work reliably)
- ✅ HA integration works

If APDS validation fails: Consider physical button fallback or accept reduced reliability. Do not ship if gestures <50% reliable.

Before VL53L4CD hold-nightlight firmware:

- ✅ VL53L4CD support verification passed or VL53L0X fallback approved
- ✅ ToF validation passed (hand-hold detection works reliably)
- ✅ HA integration works

If support verification fails and no fallback approved: Bedside ships APDS-only. Nightlight feature deferred.

If ToF validation fails: Bedside ships APDS-only. Nightlight feature deferred.

Before production firmware:

- ✅ All relevant validation checklists passed
- ✅ Integration tests passed
- ✅ Owner review and approval of validation results
- ✅ Go/no-go decision made for any deferred features

Door-side production firmware requires: ELECROW validation, door sensors, HA integration
Bedside APDS production firmware requires: APDS validation, HA integration
Bedside ToF production firmware requires: VL53L4CD verification, ToF validation, HA integration

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7. Validation Results Document

Single consolidated result document: `hardware/validation_results.md`

This document will be created during validation phases. Do not create it now.

When created, it will contain subsections for: - VL53L4CD support verification results - ELECROW board validation results - Door-side sensor validation results - Bedside sensor validation results - HA integration test results

Important: This is the output of validation work, not an input. Create it incrementally as each validation phase completes.

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8. Validation Workflow

For each validation phase: 1. Create branch: `firmware-prep/phase-N-description` 2. Create validation YAML (if needed for that phase) 3. Execute validation tests 4. Document results in `hardware/validation_results.md` 5. Commit result document 6. Create PR for owner review 7. Owner reviews and merges (or requests changes) 8. Proceed to next phase only after PR merged

One phase at a time. One branch per phase. One PR per phase.

Production door-side firmware has been written and compiles cleanly (PR #56), but hardware validation is still required before deployment. No firmware should be flashed to physical hardware until relevant validation phases pass.

Document Control

Version: 1.1
Status: Validation plan — production firmware written (PR #56), hardware validation pending
Next step: Execute ELECROW board validation (flash production firmware, verify all 20 themes)
Owner approval required: Before each phase transition